App fix F1 and staged activation — results for approval

Proposal for review (2026-11-13). Nothing here is merged into `main` or shipped in a build. The app's drug list stays at the current 14 drugs until we approve a stage.

1. What we are asked to approve

1. **App fix F1** (salt words in the recall search), with the results below.
2. **The staged activation plan** (section 6), starting with Stage 1 (68 drugs).
3. Follow-ups, each separate: **F2** (recognise IV/infusion wording such as "in NS" as non-oral), **F3** (mixed-salt products such as dextroamphetamine/amphetamine).

Decisions applied: OD-30, prioritise broad, reliable coverage (staged); OD-32, existing exclusion criteria retained unchanged until reviewed separately (digoxin stays held).

2. App fix F1

Problem. For 14 proposed drugs, the app searched FDA recall notices with the salt still attached ("hydroxyzine dihydrochloride"), because its salt-word list was incomplete. Notices that say only "hydroxyzine" were missed, which can turn into a false "No matching recall found".

Change (`mobile/src/rules/ingredients.ts`, one function):

- New salt words: bisulfate, oxalate, hydrobromide, hyclate, mononitrate, propanediol, phosphate, gluconate, palmitate, tromethamine, pamoate. "Dihydrochloride" is read as "hydrochloride" (the same products are published under both names; treating them as different would wrongly rule recalls out as "different salt").
- **Guards against conflating different drugs:**
  - A salt word never splits off a mineral or metal ("sodium phosphate", "calcium gluconate", "ferrous sulfate" and "potassium chloride" stay whole). This also fixes an existing quirk: "calcium acetate" used to be read as "calcium".
  - A new salt word only counts at the end of a name ("acidulated phosphate fluoride" stays whole).
  - Not salt words: "dinitrate" (isosorbide dinitrate is a different drug from the mononitrate), "chloride", "aspartate" and "saccharate" (mixed-salt products need F3).
- Words that were salt words before F1 are read exactly as before.

Conflation check across all of RxNorm (14,718 ingredients, 3,688 salt forms; recorded 2026-11-13 and kept as an offline test): **no ingredient name is now read as an ingredient the app checks or excludes.** 61 ingredients that used to be shortened to a metal are now kept whole; 151 salt forms now read correctly as their ingredient. 11 cosmetic or veterinary names (for example "cetyl palmitate", "tylosin phosphate") are now read as another RxNorm name; none is an oral prescription drug in scope.

Before and after, on recorded FDA responses, through the app's own evidence path (search, paging, positive control, evaluation). One real FDA product per case; extra salt forms test false positives; two controls must not change. All responses are recorded in `tests/fixtures/f1/` and replayed by `tests/integration/f1_salt_search.test.ts`.

Case	Product NDC	Records retrieved	Result for that product	New possible	New ruled out	Lost
clopidogrel bisulfate tablet	60505-0253	1 → 7	No match → No match	0	6	0
escitalopram oxalate tablet	68788-4171	0 → 2	No match → No match	0	2	0
citalopram hydrobromide tablet	0378-6231	1 → 8	No match → Possible match **(false no-match fixed)**	1	6	0
doxycycline hyclate capsule	85534-0048	17 → 30	No match → Possible match **(false no-match fixed)**	4	9	0
doxycycline monohydrate capsule	72189-358	30 → 30	Possible match → Possible match	0	0	0
isosorbide mononitrate ER tablet	68071-4290	3 → 11	No match → No match	0	8	0
dapagliflozin tablet	69097-388	0 → 1	No match → No match	0	1	0
hydroxyzine hydrochloride tablet	80425-0394	0 → 24	No match → Possible match **(false no-match fixed)**	1	23	0
hydroxyzine pamoate capsule	43063-172	4 → 24	No match → Possible match **(false no-match fixed)**	1	19	0
sitagliptin tablet	0781-5766	1 → 2	No match → No match	0	1	0
chlorhexidine gluconate oral rinse	59883-175	83 → 86	Possible match → Possible match	1	2	0
clindamycin palmitate oral solution	53746-468	2 → 50	No match → Possible match **(false no-match fixed)**	5	43	0
clindamycin hydrochloride capsule	50090-7659	50 → 50	Possible match → Possible match	0	0	0
levocetirizine tablet	82804-232	3 → 4	No match → No match	0	1	0

Case	Product NDC	Records retrieved	Result for that product	New possible	New ruled out	Lost
pramipexole tablet	29300-211	17 → 23	No match → No match	0	6	0
ketorolac tromethamine tablet	0904-7303	33 → 58	No match → No match	0	25	0
CONTROL metformin ER tablet (current drug)	62756-142	91 → 91	Possible match → Possible match	0	0	0
CONTROL potassium chloride ER tablet	60687-756	48 → 48	Possible match → Possible match	0	0	0

What this shows:

- **No false negatives introduced:** no record found before F1 is lost (last column), and no product moves toward a no-match.
- **Five false no-matches fixed:** citalopram, doxycycline hyclate, hydroxyzine (hydrochloride and pamoate) and clindamycin palmitate went from "No match" to "Possible match" because relevant recall records are now found.
- **False positives controlled:** newly found records are never "Recall match found". Records for other salts, dosage forms and strengths are ruled out: 16 hydrochloride recalls for the hydroxyzine pamoate capsule; 19 hydrochloride/phosphate recalls for the clindamycin palmitate solution. A newly found record becomes a *possible* match only when it states no comparable strength or form (for example a compounding pharmacy's "all hydroxyzine products" recall), which is the app's existing cautious rule.
- **Reviewed by hand:** 7 chlorhexidine records moved from "possible" to "ruled out": four oral-care kits (toothbrushes, swabs, solutions) and three chlorhexidine + alcohol skin swabs. Before F1 the app could not read "chlorhexidine" in these notices.
- **Known over-inclusion (F2):** one IV clindamycin infusion record ("120MG in NS ... Homepumps") is shown as a *possible* match for the oral solution, because "in NS"/"homepump" are not yet recognised as non-oral wording. It errs toward caution; F2 would fix it.
- **Controls unchanged:** metformin ER (a current drug) and potassium chloride ER retrieve and classify exactly as before.

Preserved: the four result categories, the matching rules, uncertainty handling and all patient-facing wording are unchanged (no copy or rule-table change). The full suite passes: 448 automated tests and 33 screen tests, including the build 7 scanner and summary-first result screens.

3. The 17 held entries (all still switched off)

Drug	Specific issue	Status
clopidogrel, escitalopram, citalopram, doxycycline, isosorbide mononitrate, dapagliflozin, hydroxyzine, sitagliptin, chlorhexidine, clindamycin, levocetirizine, pramipexole, ketorolac	Recall-search gap	Fixed by F1 and verified; held until F1 is approved

Drug	Specific issue	Status
dextroamphetamine/amphetamine	Mixed salts (four salts of two ingredients)	Gap remains (35 vs 37 records); needs F3
tiotropium	FDA lists its capsules as oral and inhaled	Needs a route decision
lidocaine	FDA lists viscous lidocaine as oral and topical	Needs a route decision
digoxin	Narrow therapeutic index	Held for the separate OD-32 review

4. Review of the 130 proposed active entries

Every oral prescription product in FDA's NDC Directory with exactly each drug's ingredient set was run through the app's own identity and eligibility code (`scripts/catalog/step6_product_review.ts`; results in `data/proposals/extended_drug_list_product_review.json`).

- **Product identity:** 22,380 FDA products across the 144 active drugs; 21,882 matched to the right drug; 461 are metoprolol products correctly matched to the other metoprolol salt; none matched a wrong drug.
- **Exclusion policy:** no product of any proposed drug is excluded, and none carries an anticoagulant or transplant-immunosuppressant class signal. The four excluded drugs (apixaban, rivaroxaban, warfarin, cyclosporine) remain blocked.
- **Dosage form and route:** the only cautious answers are 14 metoprolol products (counted under both metoprolol entries) where FDA's own record disagrees on the salt ("needs clarification", the known OD-13 case), 8 lactulose products listed as oral and rectal and 1 nystatin product listed as oral and topical ("not covered"). Nitroglycerin's products are mostly tongue sprays (moved to Stage 3).
- **Release type:** 38 proposed drugs have products with more than one release type (for example bupropion IR/ER); the app keeps them distinct and asks when the label does not say (Stage 2).
- **Not oral for most products:** albuterol, semaglutide, methylprednisolone, estradiol, nystatin, ketoconazole, timolol, hydrocortisone, nitroglycerin and methotrexate (Stage 3).

5. Counts and coverage, reconciled

- **Ranking:** 200 CMS generic names → 196 rows (four pairs are the same ingredient: doxycycline, hydroxyzine, ondansetron, fenofibrate; their claims are added, never double-counted). Combination products are separate CMS names with their own claims, so single-ingredient and combination rows never overlap.

- **Catalog:** 161 drugs = 14 current + 130 proposed active + 17 held; 159 ingredients (combinations share ingredients, for example hydrochlorothiazide appears in four drugs).
- **Decisions:** 14 current + 130 + 17 + 4 excluded + 31 out of scope = 196 rows.
- **Denominator:** all 1,616,932,462 Part D claims in 2023 (all drugs, all forms); 20,185,472 of the top 200's claims are vaccines and supplies. Claims count fills, Medicare only.
- **Coverage range:** an upper bound (every claim of a covered generic name, including forms the app does not check) and an oral-weighted estimate (claims × the share of the drug's FDA prescription products that are oral-only; a proxy, as CMS does not split claims by dosage form):

Scope	Drugs	Upper bound	Oral-weighted estimate
Today	14	29.1%	28.1%
+ Stage 1	+68	51.9%	50.2%
+ Stage 2	+40	65.8%	63.4%
+ Stage 3	+22	74.5%	69.3%
+ Stage 4 (held)	+17	79.2%	73.5%

6. Staged activation plan

Stage	Drugs	Criteria	Gate before switching on
Stage 1	68	Single ingredient; oral tablets/capsules; one release type; every FDA product matched to the drug; not a controlled substance; mostly oral	Recorded-data flow tests for 10 drugs; Pixel test with 3 labels; team review of the stage rows
Stage 2	40	Combination products, drugs with several release types (the app asks when the label does not say), mostly liquids, or a few FDA products answered cautiously	Same, plus release-question and combination flows; 10 more drugs on recorded data
Stage 3	22	Controlled substances; generic names whose FDA products are mostly not oral (inhalers, patches, eye drops, sprays); methotrexate (clinical review)	Team decision on how to present non-oral forms; clinical review of controlled substances and methotrexate
Stage 4	17	The 17 held: 13 fixed by F1 (after F1 approval); dextroamphetamine/amphetamine (needs F3); tiotropium and lidocaine (route data review); digoxin (OD-32 review)	Each drug only when its specific issue is closed and re-verified

Each stage: re-run the pipeline (FDA data changes), switch the stage's drugs to ACTIVE in a new catalog version, review the change summary, approve that exact catalog version (OD-2), then build and device-test before any distribution.

Stage 1 (68): hydrochlorothiazide, trazodone, simvastatin, sertraline, famotidine, duloxetine, montelukast, meloxicam, pravastatin, allopurinol, amoxicillin, azithromycin, alendronate, empagliflozin, spironolactone, mirtazapine, donepezil, ibuprofen, cephalexin, buspirone, ezetimibe, finasteride, atenolol, tizanidine, hydralazine, celecoxib, baclofen, ciprofloxacin, glimepiride, ondansetron, valsartan, aripiprazole, risperidone, olanzapine, nitrofurantoin, mirabegron, amitriptyline, fluconazole, pioglitazone, olmesartan, valacyclovir, amiodarone, chlorthalidone, meclizine, torsemide, lovastatin, methocarbamol, levofloxacin, irbesartan, hydroxychloroquine, dicyclomine, linaclotide, sucralfate, benazepril, bumetanide, enalapril, cefdinir, benztropine, icosapent_ethyl, acyclovir, sumatriptan, anastrozole, ramipril, ergocalciferol, linagliptin, promethazine, dexamethasone, colchicine

Stage 2 (40): metoprolol_tartrate, potassium_chloride, carvedilol, prednisone, quetiapine, bupropion, diclofenac, glipizide, fluoxetine, cyclobenzaprine, amoxicillin_clavulanate, diltiazem, venlafaxine, lisinopril_hydrochlorothiazide, memantine, oxybutynin, losartan_hydrochlorothiazide, valproate, levetiracetam, sulfamethoxazole_trimethoprim, lamotrigine, propranolol, carbidopa_levodopa, clonidine, nifedipine, prednisolone, paroxetine, ropinirole, topiramate, sacubitril_valsartan, naproxen, triamterene_hydrochlorothiazide, esomeprazole, doxazosin, metronidazole, fenofibrate, lactulose, amlodipine_benazepril, verapamil, valsartan_hydrochlorothiazide

Stage 3 (22): albuterol, tramadol, alprazolam, oxycodone, clonazepam, oxycodone_acetaminophen, lorazepam, semaglutide, zolpidem, pregabalin, methylprednisolone, estradiol, nystatin, ketoconazole, diazepam, timolol, hydrocortisone, morphine, acetaminophen_codeine, nitroglycerin, temazepam, methotrexate

Stage 4 (17, held): clopidogrel, escitalopram, citalopram, doxycycline, isosorbide_mononitrate, dapagliflozin, hydroxyzine, sitagliptin, chlorhexidine, tiotropium, clindamycin, dextroamphetamine_amphetamine, levocetirizine, pramipexole, digoxin, ketorolac, lidocaine

7. Not done (by request)

No merge into `main`, no new build, no distribution, no catalog activation. Everything is on branch `task-4c/scanner` for review.